EVALUATION OF NLP SYSTEMS

Lecture 2 - Paradigms

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- perspectives and approaches
- taxonomy

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WHO CARES?

users

system developers

researchers

community

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Three kinds of evaluation

Hirschman & Thompson (1997)

ADEQUACY

determination of the fitness of a system for a purpose: will it do what is required, how well, at what cost, etc.

DIAGNOSTIC

production of a system performance profile with respect to some taxonimization of the space of possible inputs

PERFORMANCE

measurement of system performance in one or more specific areas and comparison to other systems

Towards better NLP system evaluation

Spärck Jones (1994)

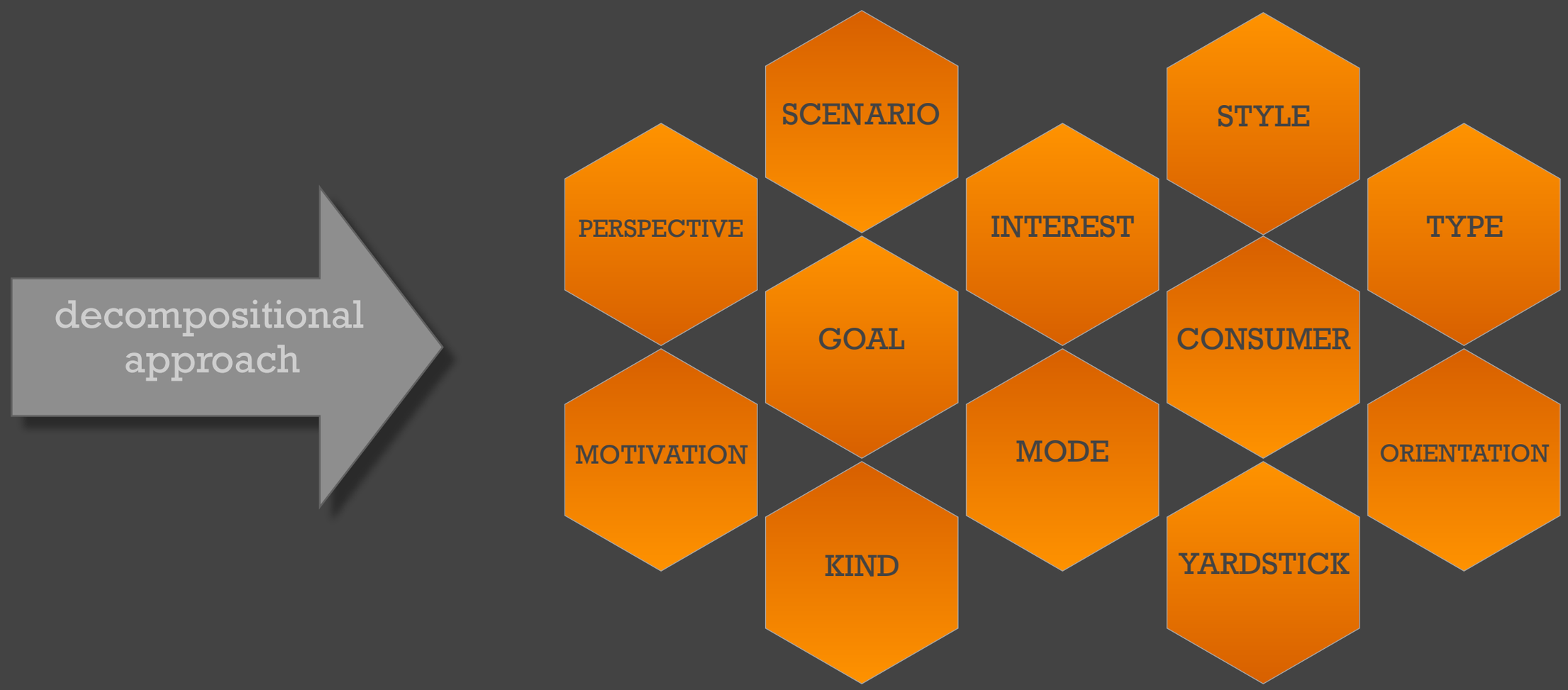
Source

"[…] there are no easy ways of evaluating systems, no magic numbers encapsulating performance, no 'core' functions that can be pursued far in isolation, no fixed meaning-representation devices any system must have."

- evaluation has to be approached comprehensively and systematically
- no single correct way to evaluate an NLP system → very different evaluations are required in different circumstances
- in all cases the context surrounding the NLP subject being evaluated matters
- four clusters of concepts:
  - evaluation subjects and performance factors
  - kinds and levels of definition for performance assessments
  - sorts of test and evaluation data
  - strategies for evaluation design and conduct

Towards better NLP system evaluation

Spärck Jones (1994)



The 7-step recipe

EAGLES (1999)

1. Why is the evaluation being done?

- what is the purpose of the evaluation? do all parties involved have the same understanding of the purpose?
- what exactly is being evaluated? is it a system or a system component? a system in isolation or a system in a specific context of use? where are the boundaries of the system?

2. Elaborate a task model

- identify all relevant roles and agents
- what is the system going to be used for?
- who will use it? what will they do with it? what are these people like?

3. Define top level quality characteristics

- what features of the system need to be evaluated? are they all equally important?

4. Produce detailed requirements for the system under evaluation, on the basis of 2 and 3

- for each feature which has been identified as important, can a valid and reliable way be found of measuring how the object being evaluated performs with respect to that feature? if not, then the features have to be broken down in a valid way, into sub-attributes which are measurable. this point has to be repeated until a point is reached where the attributes are measurable.

5. Devise the metrics to be applied to the system for the requirements produced under 4

- both measure and method for obtaining that measure have to be defined for each attribute.
- for each measurable attribute, what will count as a good score, a satisfactory score or an unsatisfactory score given the task model (2)? where are the cut off points?
- usually, an attribute has more than one sub-attributes. how are the values of the different sub-attributes combined to a value for the mother node in order to reflect their relative importance (again given the task model)?

6. Design the execution of the evaluation

- develop test materials to support the testing of the object.
- who will actually carry out the different measurements? when? in what circumstances? what form will the end result take?

7. Execute the evaluation

- make measurement.
- compare with the previously determined satisfaction ratings.
- summarize the results in an evaluation report, cf. point 1.

- [x] perspectives and approaches
- [ ] taxonomy

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10 CONTRASTS

to have in mind when evaluating an NLP system

manual

recruit human subjects and ask them to assess system output along some predetermined criteria

Resnik & Lin (2010))

automatic

evaluation algorithms that mimic the behavior of human assessors

Resnik & Lin (2010))

extrinsic

the system output is assessed in its impact on a task external to the system itself

Resnik & Lin (2010))

instrinsic

the system output is directly evaluated in terms of a set of norms or predefined criteria about the desired functionality of the system itself

Resnik & Lin (2010))

formative

inform the designer as to whether progress is being made towards the intended goals

Resnik & Lin (2010))

summative

conducted once a system is complete (or has reached a major milestone in its development)

Resnik & Lin (2010))

users

humans who use the system, not necessarily with technical knowledge

experts

humans have expertise on evaluating the system

component

evaluate each component individually

Resnik & Lin (2010))

end-to-end

consider multiple components at once

Resnik & Lin (2010))

glass box

examines the mechanisms linking input and output

Galliers & Spärck Jones (1996)

black box

considers only system input-output relations without regard to the specific mechanisms by which the outputs were obtained

Galliers & Spärck Jones (1996)

investigative

concerned with an operational system or setup and with establishing its performance characteristics

Galliers & Spärck Jones (1996)

experimental

designed to answer not the question 'what is happening?' but 'what would happen if so and so?'

Galliers & Spärck Jones (1996)

subjective

measurements are based on the perception that human beings have of such process

Paroubek et al. (2007)

objective

measurements are performed directly on data produced by the process under test

Paroubek et al. (2007)

qualitative

the result is a label descriptive of the behavior of a system

Paroubek et al. (2007)

quantitative

the result is the value of the measurement of a particular variable

Paroubek et al. (2007)

unsupervised

no true outputs to compare to (e.g. clustering)

supervised

when annotated data or gold labels are available

See you next week!

- ☑ perspectives and approaches
- ☑ taxonomy

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